

RaceDay API Endpoint Plan

Base URL: `/api`

Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	<code>/api/auth/register</code>	Register a new user account (Participant or Organiser)	Public	<pre>{ "email": "string", "password": "string", "full_name": "string", "role": "string", "date_of_birth": "date", "phone_number": "string" }</pre>	201 Created - User object with token 400 Bad Request - Validation errors 409 Conflict - Email already exists
POST	<code>/api/auth/login</code>	Authenticate user and return JWT token	Public	<pre>{ "email": "string", "password": "string" }</pre>	200 OK - <pre>{ "token": "string", "user": {user object} }</pre> 401 Unauthorized - Invalid credentials
POST	<code>/api/auth/logout</code>	Invalidate user's current session/token	Any (logged in)	None	200 OK - <pre>{ "message": "Logged out successfully" }</pre>
GET	<code>/api/auth/me</code>	Get current authenticated user's profile	Any (logged in)	None	200 OK - User object 401 Unauthorized - Not authenticated

User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/{id}	Get user profile by ID	Any (logged in)	None	200 OK - User object (limited fields for non-owners) 404 Not Found - User not found
PUT	/api/users/{id}	Update user profile	Any (logged in, own profile)	{ "full_name": "string", "date_of_birth": "date", "phone_number": "string" }	200 OK - Updated user object 403 Forbidden - Cannot update other users 404 Not Found - User not found
PUT	/api/users/{id}/password	Update user password	Any (logged in, own profile)	{ "current_password": "string", "new_password": "string" }	200 OK - { "message": "Password updated" } 400 Bad Request - Invalid current password 404 Not Found - User not found
GET	/api/users/{id}/enrolments	Get all event enrolments for a user	Any (logged in)	None	200 OK - Array of enrolment objects with event and category details 404 Not Found - User not found
GET	/api/users/{id}/results	Get all results for a user	Any (logged in)	None	200 OK - Array of result objects with event and category details 404 Not Found - User not found

Event Management Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/events</code>	Get all events with filtering options	Public	None (query params: status, datefrom, dateto, location)	200 OK - Array of event objects
GET	<code>/api/events/{id}</code>	Get detailed event information by ID	Public	None	200 OK - Event object with categories, weather, and route details 404 Not Found - Event not found
POST	<code>/api/events</code>	Create a new event	Organiser	<pre>{ "event_name": "string", "event_date": "date", "location": "string", "description": "string", "max_participants": "int", "registration_deadline": "date" }</pre>	201 Created - Event object 400 Bad Request - Validation errors 403 Forbidden - Not an organiser
PUT	<code>/api/events/{id}</code>	Update an existing event	Organiser (owner)	<pre>{ "event_name": "string", "event_date": "date", "location": "string", "description": "string", "max_participants": "int", "registration_deadline": "date", "event_status": "string" }</pre>	200 OK - Updated event object 403 Forbidden - Not the organiser 404 Not Found - Event not found
DELETE	<code>/api/events/{id}</code>	Delete an event (soft delete - mark as cancelled)	Organiser (owner)	None	200 OK - <pre>{ "message": "Event cancelled" }</pre> 403 Forbidden - Not the organiser 404 Not Found - Event not found
GET	<code>/api/events/upcoming</code>	Get all upcoming events	Public	None	200 OK - Array of upcoming event objects
GET	<code>/api/events/my-events</code>	Get all events organised by the current user	Organiser	None	200 OK - Array of event objects created by the organiser

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					403 Forbidden - Not an organiser

Category Management Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{event_id}/categories	Get all categories for a specific event	Public	None	200 OK - Array of category objects 404 Not Found - Event not found
GET	/api/categories/{id}	Get category details by ID	Public	None	200 OK - Category object 404 Not Found - Category not found
POST	/api/events/{event_id}/categories	Create a new category for an event	Organiser (event owner)	{ "category_name": "string", "distance_km": "decimal", "entry_fee": "decimal", "age_min": "int", "age_max": "int", "gender_restriction": "string", "capacity": "int", "start_time": "time" }	201 Created - Category object 403 Forbidden - Not the event organiser 404 Not Found - Event not found
PUT	/api/categories/{id}	Update an existing category	Organiser (event owner)	{ "category_name": "string", "distance_km": "decimal", "entry_fee": "decimal", "age_min": "int", "age_max": "int", "gender_restriction": "string", "capacity": "int", "start_time": "time" }	200 OK - Updated category object 403 Forbidden - Not the event organiser 404 Not Found - Category not found
DELETE	/api/categories/{id}	Delete a category	Organiser (event owner)	None	200 OK - { "message": "Category deleted" } 403 Forbidden - Not the event organiser 404 Not Found -

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					Category not found
GET	/api/events/{event_id}/categories/{category_id}/enrolments	Get all enrolments for a specific category	Organiser (event owner)	None	200 OK - Array of enrolment objects with participant details 403 Forbidden - Not the event organiser 404 Not Found - Category not found

Event Enrolment Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	<code>/api/events/{event_id}/enrol</code>	Enrol a participant in an event category	Any (logged in)	<code>{ "category_id": "int" }</code>	201 Created - Enrolment object 400 Bad Request - Category full or validation errors 404 Not Found - Event or category not found 409 Conflict - Already enrolled
GET	<code>/api/enrolments/{id}</code>	Get enrolment details by ID	Any (logged in)	None	200 OK - Enrolment object with event, category, and participant details 404 Not Found - Enrolment not found
PUT	<code>/api/enrolments/{id}</code>	Update enrolment details (e.g., change category)	Any (logged in, own enrolment)	<code>{ "category_id": "int", "status": "string", "medical_conditions": "string", "emergency_contact": "string" }</code>	200 OK - Updated enrolment object 403 Forbidden - Cannot modify others' enrolments 404 Not Found - Enrolment not found
DELETE	<code>/api/enrolments/{id}</code>	Cancel enrolment/withdraw from event	Any (logged in, own enrolment)	None	200 OK - <code>{ "message": "Enrolment cancelled" }</code> 403 Forbidden - Cannot cancel others' enrolments 404 Not Found - Enrolment not found
GET	<code>/api/events/{event_id}/enrolments</code>	Get all enrolments for an event	Organiser (event owner)	None	200 OK - Array of enrolment objects with participant and category details 403 Forbidden - Not the event organiser

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					404 Not Found - Event not found
GET	<code>/api/events/{event_id}/enrolments/statistics</code>	Get enrolment statistics for an event	Organiser (event owner)	None	200 OK - Statistics object <code>{ "total": "int", "by_category": {}, "status_counts": {} }</code> 403 Forbidden - Not the event organiser 404 Not Found - Event not found
GET	<code>/api/me/enrolments</code>	Get all enrolments for the current user	Any (logged in)	None	200 OK - Array of enrolment objects with event and category details

Results Management Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	<code>/api/enrolments/{enrolment_id}/results</code>	Capture/upload results for a participant	Organiser (event owner)	<code>{ "finish_time": "time", "gun_time": "time", "chip_time": "time", "status": "string" }</code>	201 Created - Result object 403 Forbidden - Not the event organiser 404 Not Found - Enrolment not found 409 Conflict - Result already exists
GET	<code>/api/results/{id}</code>	Get result details by ID	Public	None	200 OK - Result object with enrolment and participant details 404 Not Found - Result not found
PUT	<code>/api/results/{id}</code>	Update/verify a result	Organiser (event owner)	<code>{ "finish_time": "time", "gun_time": "time", "chip_time": "time", "overall_position": "int", "category_position": "int", "gender_position": "int", "pace_per_km": "decimal", "status": "string" }</code>	200 OK - Updated result object 403 Forbidden - Not the event organiser 404 Not Found - Result not found
DELETE	<code>/api/results/{id}</code>	Delete/void a result	Organiser (event owner)	None	200 OK - <code>{ "message": "Result voided" }</code> 403 Forbidden - Not the event organiser 404 Not Found - Result not found
GET	<code>/api/events/{event_id}</code>	Get all results for an event	Public	None (query params: category_id)	200 OK - Array of result objects

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
	<code>/api/events/{event_id}/results</code>				with participant details 404 Not Found - Event not found
GET	<code>/api/events/{event_id}/results/leaderboard</code>	Get leaderboard for an event	Public	None (query params: category_id)	200 OK - Array of ranked results with positions 404 Not Found - Event not found
GET	<code>/api/events/{event_id}/results/download</code>	Download all results as CSV/PDF	Organiser (event owner)	None	200 OK - CSV/PDF file 403 Forbidden - Not the event organiser 404 Not Found - Event not found
POST	<code>/api/events/{event_id}/results/bulk</code>	Bulk upload multiple results	Organiser (event owner)	<pre>{ "results": [{ "enrolment_id": "int", "finish_time": "time", "position": "int" }] }</pre>	201 Created - Bulk upload summary 400 Bad Request - Invalid data format 403 Forbidden - Not the event organiser 404 Not Found - Event not found

Weather Information Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/events/{event_id}/weather</code>	Get weather forecast for an event	Public	None	200 OK - Weather object or array 404 Not Found - Event not found
POST	<code>/api/events/{event_id}/weather</code>	Add/update weather forecast	Organiser (event owner)	<pre>{ "temperature": "decimal", "conditions": "string", "humidity": "int", "wind_speed": "decimal", "rainfall_mm": "decimal", "weather_source": "string" }</pre>	201 Created - Weather object 400 Bad Request - Validation errors 403 Forbidden - Not the event organiser 404 Not Found - Event not found
PUT	<code>/api/weather/{id}</code>	Update weather information	Organiser (event owner)	<pre>{ "temperature": "decimal", "conditions": "string", "humidity": "int", "wind_speed": "decimal", "rainfall_mm": "decimal" }</pre>	200 OK - Updated weather object 403 Forbidden - Not the event organiser 404 Not Found - Weather record not found
GET	<code>/api/events/{event_id}/weather/forecast</code>	Get 5-day forecast for event location	Public	None	200 OK - Array of forecast objects 404 Not Found - Event not found

Route Information Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{event_id}/routes	Get all route information for an event	Public	None	200 OK - Array of route objects 404 Not Found - Event not found
GET	/api/routes/{id}	Get specific route details	Public	None	200 OK - Route object 404 Not Found - Route not found
POST	/api/events/{event_id}/routes	Add route information to an event	Organiser (event owner)	{ "route_name": "string", "distance_km": "decimal", "elevation_gain": "int", "elevation_loss": "int", "route_description": "string", "gpx_file_url": "string", "water_points": "int", "aid_stations": "int" }	201 Created - Route object 403 Forbidden - Not the event organiser 404 Not Found - Event not found
PUT	/api/routes/{id}	Update route information	Organiser (event owner)	{ "route_name": "string", "distance_km": "decimal", "elevation_gain": "int", "elevation_loss": "int", "route_description": "string", "gpx_file_url": "string", "water_points": "int", "aid_stations": "int" }	200 OK - Updated route object 403 Forbidden - Not the event organiser 404 Not Found - Route not found
DELETE	/api/routes/{id}	Delete route information	Organiser (event owner)	None	200 OK - { "message": "Route deleted" } 403 Forbidden - Not the event organiser 404 Not Found - Route not found

Dashboard & Statistics Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/dashboard/organiser	Get organiser dashboard statistics	Organiser	None	200 OK - { "total_events": "int", "total_participants": "int", "upcoming_events": "int", "revenue": "decimal", "recent_activity": [] } 403 Forbidden - Not an organiser
GET	/api/dashboard/participant	Get participant dashboard statistics	Any (logged in)	None	200 OK - { "total_events_entered": "int", "upcoming_events": "int", "completed_events": "int", "personal_bests": {}, "recent_results": [] } 404 Not Found - Event not found
GET	/api/events/{event_id}/statistics	Get comprehensive statistics for an event	Public	None	200 OK - { "total_participants": "int", "by_category": {}, "gender_breakdown": {}, "age_breakdown": {}, "average_time": "time", "fastest_time": "time" } 404 Not Found - Event not found

Endpoint Summary Statistics

Category	Count
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Category Management	6
Event Enrolment	6
Results Management	8
Weather Information	4
Route Information	5
Dashboard & Statistics	3
TOTAL	48

Role-Based Access Summary

Role	Allowed Operations
Public (No Auth)	View events, categories, results, weather, routes
Participant (Authenticated)	Register/login, view own profile, enrol in events, view own enrolments and results, update own profile
Organiser (Authenticated)	All Participant permissions + Create/update/delete events, manage categories, capture results, view all event enrolments, generate reports

Notes & Conventions

Authentication

- All endpoints except Public ones require a valid JWT token
- Token must be included in the `Authorization: Bearer {token}` header

Pagination

- GET endpoints with list responses support pagination via `?page={page}&limit={limit}`
- Default: `page=1`, `limit=20`

Filtering Parameters

Parameter	Description	Example
<code>status</code>	Filter by status	<code>?status=Upcoming</code>
<code>category_id</code>	Filter by category	<code>?category_id=1</code>
<code>date_from</code>	Start date filter	<code>?date_from=2026-01-01</code>
<code>date_to</code>	End date filter	<code>?date_to=2026-12-31</code>
<code>search</code>	Text search	<code>?search=comrades</code>
<code>sort_by</code>	Sort field	<code>?sort_by=event_date</code>
<code>sort_order</code>	Sort direction (ASC/DESC)	<code>?sort_order=DESC</code>

Status Codes

Code	Meaning
200	OK - Successful GET/PUT/DELETE
201	Created - Successful POST
400	Bad Request - Validation errors
401	Unauthorized - Authentication required
403	Forbidden - Insufficient permissions
404	Not Found - Resource doesn't exist
409	Conflict - Resource conflict (duplicate, etc.)

Error Response Format

```json { "error": {

```
"code": "VALIDATION_ERROR",
 "message": "Invalid input data",
 "details": {
 "field": "email",
 "issue": "must be a valid email address"
 }
}
```

}}